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The influence of purchase situation on buying center structure and involvement: a select meta-analysis of organizational buying behavior research

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Abstract

Researchers investigating various aspects of organizational buying behavior often have reported mixed, sometimes contradictory results. Recently, there have been attempts to consolidate work in this area [cf. Bunn, M.D. 1993. Taxonomy of buying decision approaches. Journal of Marketing; 57:38–56 (January); Johnston, W.J., Lewin, J.E. Organizational buying behavior: toward an integrative framework. Journal of Business Research 1996; 35 (January): 1–15]. However, to date no one has attempted a meta-analytical integration of this research stream. To fill this gap, the authors conduct a meta-analysis of the relationships between the nature of the purchase situation and buying center structure and buying center involvement. While results indicate that cumulative findings in some cases are robust, in other cases they are not. Subsequent moderator analysis indicates that study design characteristics account for significant variation in findings across studies.
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1. Introduction

Over the past decades, researchers have studied the processes and behaviors used by organizations in their purchasing activities. Marketers, in particular, have encouraged these investigations as an aid to better understand, serve, and retain their organizational customers. As a result, scores of theoretical and empirical articles have examined the constructs associated with organizational buying behavior.

Additionally, like most business activities, the environment surrounding organizational purchasing is dynamic—evolving with the emergence of new techniques and technologies (Cannon and Perrault, 1999). For example, in recent years new processes have been introduced to aid organizations in their procurement efforts. These aids include (1) Web catalogs offering product specification, price, and availability, (2) Internet-based ordering and tracking systems, (3) electronic data interchange to facilitate inventory control, credit approval, invoicing and receivables, and (4) direct communication and relationship management tools. These new activities and aids provide opportunities for new research in organizational purchasing. However, before conducting new research a meta-analysis of the existing literature could prove useful in several ways. For example, by quantitatively summarizing the accumulated body of knowledge systematic sources of differences can be identified (Farley and Lehmann, 1986), new work in the field can be more effectively designed (Farley et al., 1998), and certain unresolved questions may be either clarified or settled (Brown and Peterson, 1993).

One of these unresolved questions is elucidated by McQuiston (1989) who points out that researchers have had mixed success examining participation and influence

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(involvement) in organizational purchasing. Additionally, McCabe (1987) observes that the literature continues to support apparently conflicting views of the relation between buying center structure and the nature of the purchase situation. Summarizing these issues, Ghingold and Wilson (1998) argue that generalizable conclusions regarding the dynamic nature of buying center structure and its implications remain cloudy at best.

To fill this void and to address some of the inconsistencies currently existing in the literature, we conduct a *select* meta-analytical integration of organizational buying behavior research. More specifically, we examine the strength, significance, and generality of relationships between the nature of the *purchase situation* and *(a) buying center structure* and *(b) buying center involvement* (see Fig. 1). We also examine several study design characteristics that may moderate the relationships between these antecedent and outcome variables.

The work reported here makes several specific contributions. First, we provide a systematic investigation of the extant literature examining the relationship between the nature of the purchase situation and buying center structure and involvement. Second, we identify unresolved questions within the literature stream and provide some clarification to these disputed topics. Third, we add to this body of knowledge by identifying and investigating seven study design characteristics as potential moderators of the effect sizes found across studies. Our results should help both academic and industry researchers better understand how these design characteristics can influence their studies (cf. Lynch, 1999).

The remainder of this article is organized in the following manner. First, we define the *research domain*. Second, we provide a *summary of inconsistencies* found in prior research. Third, we describe our investigative *procedures*. Fourth, we discuss the *results* of our analyses. Finally, we review the *implications* and *limitations* of our study.

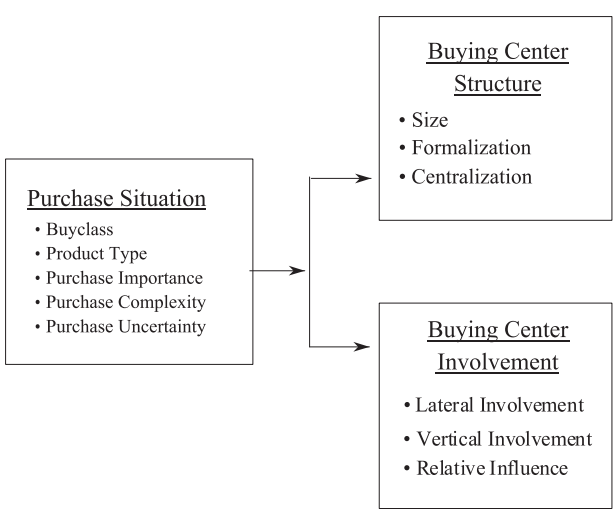

Fig. 1. The meta-analysis research domain.

2. Research domain

2.1. Antecedent variables: purchase situation

Across the empirical work reviewed for this meta-analytic effort, the nature of the 'purchase situation' was the most frequently examined antecedent construct found (79.5%). The distant second and third frequently examined antecedents were organizational size (11.8%) and buyphase (8.7%). Because of this—and because of the relatively small respective cell sizes related to these latter two constructs—we decided to focus the effort in this study on the group of antecedent constructs characterizing the *purchase situation*.

In our review, five purchase specific constructs were used repeatedly (but with mixed success) to represent the type of purchase situation under consideration: (1) buyclass, (2) product type, (3) purchase importance, (4) purchase complexity, and (5) purchase uncertainty. It is interesting to note that throughout the literature these five situation constructs often are compared and intertwined. Select examples of this are included in the following discussion.

2.1.1. Buyclass

First, numerous studies have examined the influence buyclass (i.e., new task, modified rebuy, straight rebuy) has on buying center structure and involvement (e.g., Reese and Stone, 1987; McWilliams et al., 1992). Many of these studies suggest that "organizational buyers regard new tasks as important" (e.g., Dholakia et al., 1993, p. 283), and "new tasks are perceived as high risk" (e.g., Anderson et al., 1987, p. 72). Additional research characterize new task purchases as being more complex (e.g., Lund, 1989), and associated with greater uncertainty (e.g., Bunn, 1993). In contrast, rebuy situations frequently are characterized in the literature as being less complex and uncertain, and associated with less risk (e.g., Mitchell, 1998; Jackson et al., 1984).

2.1.2. Product type

Second, the type of product involved in a purchase situation also has been used in studies to explain variance in buying center structure and involvement. Over the years the product type schema has been expanded to include seven categories ranging from major capital equipment to business services (cf. Johnston and Bonoma, 1981; Lilien and Wong, 1984). Some researchers have proposed a commonality between buyclass and product type. For example, Bellizi and McVey (1983) observe that there may exist a conceptual overlap between product type and buyclass. They go on to compare major capital equipment and major materials with a new task purchase situation, minor capital equipment and minor materials with a modified rebuy situation, and routine operating supplies with a straight rebuy purchase situation.

2.1.3. Purchase importance and purchase complexity

In the literature, purchase importance frequently is defined as the relative importance of the current purchase to other purchases of similar type and/or the current purchase's perceived impact on the organization (e.g., Dawes et al., 1993; Stump and Heide, 1996). Purchase complexity often is defined as the technical complexity of the product and/or the complexity of the buying decision or task under consideration (cf. McCabe, 1987; Lewin and Bello, 1997). Interestingly, several authors have criticized the buyclass model for failing to include the importance and/or complexity of the purchase under consideration (cf. Johnston and Bonoma, 1981; McQuiston, 1989).

2.1.4. Purchase uncertainty

The impact of purchase uncertainty also has been used to explain variations in buying center structure and involvement (e.g., Spekman and Stern, 1979; Lewin and Bello, 1997). Purchase uncertainty is often characterized as the uncertainty regarding the information available to make the best decision and/or the ultimate positive (negative) outcome of a purchase decision. Here too, similarities and overlap between purchase uncertainty and other purchase-related constructs have been proposed. For example, Morris et al. (1995) observe that uncertainty is a key characteristic in several approaches to classifying industrial buying. While, Dawes et al. (1993) argue that task uncertainty is closely intertwined with the buygrid dimension of buyclass.

2.1.5. Purchase situation: the interactive/combined effect

As cited in the prior paragraphs, these five purchase situation variables often are discussed in relation to one another throughout the organizational buying behavior (OBB) literature. For example, the three buyclass categorizations are frequently described in terms of their levels of importance, complexity, and/or uncertainty. In turn, important, complex, and/or uncertain purchase situations often are described as 'new task' in nature. Similar comparisons are found within discussions surrounding the various product type classifications.

Because of these suggested interactions, we create a summary construct, 'purchase situation' to systematically analyze the combined effect of the individual purchase situation constructs—buyclass, product type, purchase importance, complexity, and uncertainty—on the focal outcome variables. Our approach to creating this summary construct will be discussed in Section 5. In the next sections we will define the outcome variables examined in our meta-analysis, and provide a summary of inconsistencies across prior studies.

2.2. Outcome variables: buying center structure and involvement

While many purchase-related outcome variables have been studied over the years, six emerge as those most often examined: These six outcome variables are (1) buying center size, (2) formalization, (3) centralization, (4) lateral involvement, (5) vertical involvement, and (6) the relative influence of various buying center participants. Buying center size, along with both formalization and centralization in purchase decision making, often are referred to as components of buying center structure (cf. Spekman and Stern, 1979; McCabe, 1987). Whereas, both lateral and vertical involvement in purchase decision making, along with the relative influence of buying center participants, often are characterized as components of buying center involvement (cf. Jackson et al., 1984; McQuiston, 1989).

2.2.1. Buying center structure: size, formalization, and centralization

Buying center *size* refers to the number of people actively involved in a purchase situation across the various stages of the decision process. Currently, many accept the notion that buying centers will be larger when the purchase decision is of significant importance and/or the purchase situation either is unfamiliar or complex. In contrast, smaller buying centers are expected when the purchase decision is of lesser importance, is more familiar, and/or more routine.

Buying center *formalization* refers to the emphasis placed on the use of formal rules and procedures by buying center participants during the purchase process. In more formalized buying centers little, if any, flexibility is expected beyond what is set forth by policy within the customer organization. In these cases, understanding the vendor's formal guidelines are key aspects of success. Buying center *centralization* refers to buying centers where, regardless of the number of people involved, only a few participants hold meaningful influence over the purchase decision process. In highly centralized buying centers vendors must ascertain who the key decision makers are and focus their efforts on those 'central' individuals. In contrast, in buying centers characterized by less centralization vendors must direct their efforts toward a wider array of buying center members and constituencies.

2.2.2. Buying center involvement: lateral and vertical involvement, and relative influence

Lateral involvement refers to the number of departments or other work-related groups that are represented in the buying center or are involved in some way in the purchase decision. As such, determining lateral involvement, more than simply determining the size of the buying center, seeks to gauge what will be the most important factors considered (based on those involved in decision making) in choosing a particular vendor. *Vertical involvement* refers to the levels of management that will be involved in and/or influence a particular purchase decision. Vertical involvement differs from buying center centralization in that in highly centralized buying centers only a few members have significant influence over the purchase decision. In contrast, in buying centers com-

prised of several management levels influence may be equally shared by participants. Like lateral involvement, the purpose of determining vertical involvement is the identification of all buying members who will influence the decision, so that marketing efforts may be directed toward all appropriate participants. The *relative influence* of buying center participants takes involvement a step further. In attempting to ascertain relative influence, researchers try to predict those purchase situations where, for example, the purchasing department will take a decisive role, as opposed to those purchase situations where end users, technical specialists, or management will take a lead role.

3. Summary of inconsistencies among studies

Grounded in decades of theoretical development and empirical research, the relationships among the type of purchase situation and buying center structure and involvement have, in many cases, developed a sort of equilibrium. However, careful review of supporting studies reveals a variety of inconsistencies (see Fig. 2). In the following paragraphs we provide examples of these inconsistencies.

As previously mentioned, many believe buying center *size* will be greater in purchase situations of high importance, significant complexity, and/or greater uncertainty. These types of purchases likely would be characterized as 'new task' and would include purchases of *major* equipment, *major* materials, and *key* component parts. In contrast, smaller buying centers would be expected as levels of importance, complexity, and uncertainty diminish. However, empirical results supporting these viewpoints are mixed. For example, Johnston and Bonoma (1981) found a positive relationship between buying center size and purchase complexity, but Mcquiston (1989) failed to find support for this relationship. Similarly, McWilliams et al. (1992) found buying center size larger in new task purchase situations; however, Johnston and Bonoma (1981) failed to find support for this proposed relationship.

With regard to buying center *formalization and centralization* there currently exists two divergent schools of thought. The "contingency" perspective (cf. Spekman and Stern, 1979) proposes that in uncertain situations organizations will adopt more flexible (organic) structures. Those who support this perspective predict that buying centers involved in uncertain or novel purchase situations will be characterized by less formalization and less centralization in decision making. In contrast, the "constriction of authority" perspective (cf. McCabe, 1987; Lewin, 2001) posits that in uncertain situations organizations will adopt more rigid (mechanistic) structures. Those who support this perspective predict that buying centers involved in uncertain or novel purchase situations will be characterized by greater formalization and greater centralization in decision making.

Antecedent	Outcome	Results	Example References
Buyclass	Buying Center Size	supported	Dholakia et al. 1993
		not supported	Johnston and Bonoma 1981
Buyclass	Vertical Involvement	supported	Reese and Stone 1987
		not supported	Johnston and Bonoma 1981
Buyclass	Relative Influence	supported	Pingry 1974 Naumann, Lincoln and McWilliams 1984
		not supported	Bellizi and McVey 1983 Jackson, Keith, and Burdick 1984
Purchase Complexity	Buying Center Size	supported	Johnston and Bonoma 1981
		not supported	McQuiston 1989
Purchase Complexity	Buying Center Centralization	Positive – supported	McCabe 1987
		Negative – supported	Lau, Goh, and Phau 1999
Purchase Uncertainty	Buying Center Centralization	supported	McCabe 1987
		not supported	Spekman and Stern 1979

Fig. 2. Select examples of conflicting results.

Further, results from studies examining the effect of uncertainty on buying center structure have been mixed. For example, Spekman and Stern (1979) hypothesized a negative relationship between purchase-related uncertainty and buying center formalization and centralization; but failed to find empirical support for these relationships. In contrast, McCabe (1987) proposed a positive relationship between purchase-related uncertainties and buying center formalization and centralization. McCabe's (1987) results did indicate a positive relationship between uncertainty and centralization; however, his results indicated a negative relationship between uncertainty and formalization.

Among the three involvement constructs, the greatest inconsistency across findings surrounds the impact purchase situation has on the *relative influence* of buying center members. For example, Naumann et al. (1984) along with Pingry (1974) found support for a positive relationship between buyclass and relative influence within buying centers. In contrast, other researchers failed to find support for this relationship (cf. Bellizi and McVey, 1983; Jackson et al., 1984). Additionally, while Spekman and Stern (1979) found a significant, positive relationship between purchase uncertainty and relative influence, McCabe (1987) found a significant, negative relationship between these two constructs—and McQuiston (1989) found no statistically significant relationship between these constructs. In the same study, McQuiston (1989) did find a significant positive relationship between purchase complexity and relative influence. But, in contrast, Spekman and Stern (1979) found a significant, negative relationship between purchase complexity and relative influence.

Inconsistencies also exist across findings surrounding vertical *involvement* in buying centers. For example, Reese and Stone (1987) found a statistically significant relationship between buyclass and vertical involvement, while Johnston and Bonoma (1981) found no statistically significant relationship between buyclass and vertical involvement. In the same study, however, Johnston and Bonoma (1981) found a significant, positive relationship between purchase complexity and vertical involvement, whereas McCabe (1987) found a significant negative relationship between purchase complexity and vertical involvement.

In light of the prior discussion, it seems plausible to suspect that research design artifacts might explain some of the inconsistencies in findings across the studies included in this meta-analysis. A discussion of these potential moderators is included in the following section.

4. Possible moderators

As is common in much of the literature, OBB researchers have utilized a diversity of study designs. In prior meta-analyses, research design artifacts have been shown to moderate findings across studies (e.g., Brown et al., 1998). In OBB research, such design variations may lead to differing assessments of buying center structure and involvement. Therefore, we examined seven study/design characteristics as potential moderators, as discussed next.

4.1. Sample sizes and response rates

Across the studies included in this meta-analysis, sample sizes ranged from less than 50 to greater than 500, and usable response rates ranged from 10% (mail survey) to 100% (personal interviews). Inappropriate sample sizes can influence the accuracy of findings, and inadequate response rates may result in increased sampling error and non-response bias. As such, it seems tenable to posit that notable variance in sample sizes and response rates may significantly moderate the discrete findings among the studies in this meta-analysis.

4.2. Number of firms and number of industries represented

Similarly, the number of firms represented by usable responses varied significantly—from a single firm to hundreds of firms. Also, more than one-third of the studies were limited to only one or two industries (coded 0), while the balance represented numerous industries (coded 1). Both the number of firms and industries represented in studies can influence the representativeness of the sample, as well as the generalizability of the findings. In those studies that examined relatively few firms and industries, this often was recognized as a potential limitation.

4.3. Type of respondents

A significant number of studies in the OBB literature focus exclusively on purchasing professionals (coded 1) as key informants. However, the problems surrounding the use of key informant data have been discussed throughout the literature (cf. Phillips, 1981). Other studies have used a mix of functional groups (coded 0) as respondents including purchasing, engineering, manufacturing, administrative, and others. While intuitively this might appear to be a superior approach, some research points to the potential for a lack of consistency in responses among various functions (cf. Silk and Kalwani, 1982).

4.4. Data collection methods

There were two types of data collection methods employed across the studies examined in this work: mail surveys (coded 1) and personal interviews (coded 0). Both of these types of data collection methods are recognized as having both advantages and disadvantages. For example, the advantages associated with interviewing include the ability to better qualify respondents and the capacity for clarification thus reducing misunderstanding. In contrast, the advantages associated with mail surveys include greater perceived anonymity, absence of interviewer evaluation

apprehension, and the likelihood of larger sample sizes increasing external validity. The disadvantages of these methods are generally defined in terms of the advantages of the other.

4.5. Year published

Over the years, there have been notable changes in the environment surrounding organizational purchasing. For example, the relatively recent phenomena of "downsizing" and increased emphasis on "outsourcing" have altered many aspects of business organizations. Similar changes have occurred as a result of the cyclical focus in the 1970s on cost cutting, in the 1980s on total quality management, and in the 1990s on customer satisfaction and relationships. Given these fluid environments, study constructs may be subject to temporal variations when sufficient time has elapsed.

5. Method and findings

Studies for inclusion in the meta-analysis were identified by searching the PROQUEST, EBSCOHOST, INFOTRAC, ERIC, and PSYCINFO databases. Additionally, issue-by-issue searches (1967–2002) were conducted of *JBR*, JMR, JM, JAMS, *JB2B*, *JBIM*, and *IMM*. Also, references appearing in relevant publications were searched for additional studies that might be included in the analysis. Finally, we contacted more than 30 OBB researchers to identify unpublished working papers. Initially, 67 empirical works were identified as potential contributors to our meta-analysis. However, in many cases specific pairwise relationships were not directly comparable or operatationalization and/or measurement inconsistencies precluded inclusion. As a result, 33 empirical works (published between 1979 and 1999) were identified as contributors to the meta-analysis. The antecedent and outcome variables previously described were coded independently by one of the authors and two additional coders. These independent assessments resulted in an initial agreement of 93.5%. Inconsistencies were then reexamined until complete agreement was reached.

5.1. Analysis of pairwise relationships

The effect size metric used in our study was r, the zero-order Pearson product-moment correlation coefficient. Hedges and Olkin (1985) and Hunter and Schmidt (1990) recommend using the correlation coefficient, as it is easy to interpret and scale free. While most studies reported the correlation coefficient r, some reported Student's t, F-ratio, or chi-square statistics. In these cases, the t, F, or chi-square were converted to r as suggested by Hunter and Schmidt (1990). Next, the correlation coefficient for each relationship was corrected for measurement error by dividing the correlation coefficient by the product of the square root of the reliabilities of the two variables under consideration. They were further weighted by their relative sample size. For studies where the reliabilities were not reported, the weighted mean reliability across all studies was used.

In each case, the 'purchase situation' summary construct was created by calculating the simple mean of the individual applicable purchase situation constructs, and then averaging the resulting means. This approach minimized the potential biasing results in favor of large sample size studies by weighting multiple effects from the same study by the sample size (Brown and Peterson, 1993).

5.2. Analysis of moderator effects

Following Hunter and Schmidt (1990), we examined the ratio of sampling error to total variance around the weighted mean *corrected* correlations. Where that ratio is less than 75% and the chi-square statistic is statistically significant, moderator analysis is recommended. Using these guidelines, we sought to conduct moderator analyses by using bootstrapping to regress all of the antecedent-to-outcome variable correlations against the seven coded study characteristics. However, due to limited cell sizes between the individual antecedent (i.e., buyclass, product type) and outcome (i.e., centralization, lateral involvement) variables, moderator analysis was constrained to the correlations between the *summary* 'purchase situation' variable and the six outcome variables, regressed against each of the coded study design characteristics.

After first weighting each observation by its sample size, we drew 200 bootstrap samples with replacement and then ran the regression for each bootstrap sample. We then examined the proportion of the parameter estimates from the 200 regressions that had the same sign to determine statistical significance. If, for example, more than 95% of the parameter estimates were positive, that parameter was considered positive and statistically significant ($p<.05$). The median parameter estimate across the 200 regressions served as the point estimate.

5.3. Results of the meta-analysis

Table 1 presents the results of the analysis of summary categories and specific pairwise relationships. In Table 1 we report the number of study effects k, cumulative sample size N, observed correlation r, observed correlation corrected for measurement error r', and the 95% confidence intervals for the observed and corrected correlations. Relationships that do not have zero in the confidence interval are considered to be significant.

As shown in Table 1, the relationships among the predictor variables and the outcome variables (a) buying center size, (b) lateral involvement, (c) vertical involvement, and (d) relative influence all lack zero in their respective

Table 1
Analysis of summary categories and specific pairwise relationships

Relationships	Number of study effects	Cumulative N	Observed r	Corrected r'	95% Confidence interval (observed)	95% Confidence interval (corrected)
Buying center size						
Purchase situation→	14[a]	3375[a]	.31	.40	$.20 \le \rho \le .41$	$.29 \le \rho \le .50$
Buyclass→	8	2295	.35	.47	$.18 \le \rho \le .52$	$.29 \le \rho \le .64$
Purchase importance→	4	727	.23	.29	$.07 \le \rho \le .38$	$.13 \le \rho \le .44$
Buying center formalization						
Purchase situation→	10	1684	.008	.009	$-.11 \le \rho \le .13$	$-.11 \le \rho \le .13$
Purchase importance→	2	710	.03	.03	$-.33 \le \rho \le .38$	$-.33 \le \rho \le .39$
Purchase complexity→	2	142	−.03	−.04	$-.18 \le \rho \le .11$	$-.19 \le \rho \le .10$
Purchase uncertainty→	5	758	−.05	−.07	$-.22 \le \rho \le .11$	$-.23 \le \rho \le .11$
Buying center centralization						
Purchase situation→	8	1390	.12	.15	$.03 \le \rho \le .22$	$.06 \le \rho \le .25$
Purchase importance→	2	710	.03	.03	$-.15 \le \rho \le .20$	$-.15 \le \rho \le .20$
Purchase complexity→	2	142	.21	.26	$-.13 \le \rho \le .55$	$-.08 \le \rho \le .60$
Purchase uncertainty→	3	464	.09	.12	$-.01 \le \rho \le .20$	$.01 \le \rho \le .22$
Lateral involvement						
Purchase situation→	27	6160	.25	.33	$.19 \le \rho \le .32$	$.27 \le \rho \le .40$
Buyclass→	8	1355	.19	.26	$.08 \le \rho \le .30$	$.15 \le \rho \le .37$
Product type→	4	2685	.22	.28	$.13 \le \rho \le .30$	$.20 \le \rho \le .36$
Purchase importance→	7	902	.33	.43	$.20 \le \rho \le .46$	$.30 \le \rho \le .56$
Purchase complexity→	2	423	.08	.11	$.03 \le \rho \le .13$	$.06 \le \rho \le .15$
Purchase uncertainty→	6	795	.32	.43	$.17 \le \rho \le .48$	$.27 \le \rho \le .56$
Vertical involvement						
Purchase situation→	17	5029	.32	.41	$.23 \le \rho \le .41$	$.32 \le \rho \le .50$
Buyclass→	6	916	.19	.26	$.13 \le \rho \le .26$	$.19 \le \rho \le .33$
Purchase importance→	7	1283	.44	.56	$.27 \le \rho \le .62$	$.39 \le \rho \le .73$
Purchase uncertainty→	2	438	.18	.24	$.08 \le \rho \le .29$	$.13 \le \rho \le .35$
Relative influence of participants						
Purchase situation→	16	2829	.22	.29	$.15 \le \rho \le .30$	$.22 \le \rho \le .37$
Buyclass→	5	1056	.21	.29	$.11 \le \rho \le .32$	$.18 \le \rho \le .40$
Product type→	3	534	.17	.22	$.12 \le \rho \le .22$	$.17 \le \rho \le .27$
Purchase importance→	2	319	.49	.62	$-.06 \le \rho \le 1.03$	$.08 \le \rho \le 1.17$
Purchase complexity→	2	250	.16	.22	$.09 \le \rho \le .24$	$.14 \le \rho \le .30$
Purchase uncertainty→	4	670	.18	.23	$.09 \le \rho \le .27$	$.15 \le \rho \le .32$

[a] The aggregate 'Number of Study effects' and 'Cumulative N' for individual variables within a category do not always equal those of 'Purchase situation' as variables examined in only one study were included in the 'Purchase situation' analysis but are not included as a row in the Table.

corrected confidence intervals and, therefore can be considered significant. Also, each of the *corrected* correlations are positive (ranging from .11 to .47), indicating that as the various components of 'purchase situation' increase (i.e., importance, uncertainty), marketers can expect to find larger buying centers exhibiting increased levels of lateral involvement, vertical involvement, and relative influence of purchase participants. In contrast, the relationships between the predictor variables and buying center formalization are not significant, with weak average corrected correlations ranging from .009 to −.07. Similarly, the relationships between the predictor variables and buying center centralization generally are not significant—as two of the corrected confidence intervals contain zero, while a third exhibits a lower boundary of .01.

Also, upon closer scrutiny, some interesting patterns emerge. For example, purchase importance exhibits the strongest corrected correlation with the outcome variables lateral involvement (.43), vertical involvement (.56), and relative influence (.62). In contrast, buyclass, purchase complexity and product type exhibit weaker (albeit still significant) relationships with lateral involvement, vertical involvement, and relative influence. Additional implications of these findings will be presented in Section 6.

5.4. Results of the moderator regression analysis

Using the bootstrapping method previously discussed, we regressed the purchase situation–outcome variable correlations against the seven coded study characteristics.

Table 2
Results of bootstrapped moderator regression

Predictors (study characteristics)	Dependent variables (correlation between purchase situation and →)					
	Buying center size	Buying center formalization	Buying center centralization	Lateral involvement	Vertical involvement	Relative influence
Sample size	−1.53*	−.23*	−.10	−3.07*	−.92*	−5.50*
Response rate	.006	−.19	−.15	.24	−.70*	1.02*
# of firms	1.58*	.07	.107	3.21*	.72*	2.26*
# of industries[a]	.36*	.64*	−.74*	.18	.07	7.72*
Respondents[b]	.33*	−.42*	.13	.15	na[c]	−2.99*
Data collection[d]	−.91*	.15	−.83*	.005	−1.11*	1.90*
Year published	.31*	.10	.18	.26*	.28*	−5.28*
Median R^2	.51	.49	.59	.43	.42	.87

[a] 0=one or two; 1=more than two.
[b] 0=multiple functions; 1=purchasing only.
[c] Among the studies examining the effects of purchase situation on vertical involvement, none were limited to purchasing respondents only. In other words, all of these studies interviewed or surveyed multi-functional respondents.
[d] 0=personal interview; 1=mail survey.
* Indicates $p<.05$.

The results are shown in Table 2. In each regression model the coded study characteristics account for a significant amount of variance (median R^2) in effect size; ranging from 42% for vertical involvement to 87% for relative influence of participants. Thus indicating that these seven study characteristics have substantial ability to explain the between study variance in the effect of purchase situation on buying center structure and involvement.

When examining, for instance, the final regression model (purchase situation/relative influence) we see that all seven study characteristics are statistically significant ($p<.05$) and each manifest a non-extraneous beta coefficient (β's ranging from −5.50 to 7.72); combining to account for a majority of the variance (median R^2=.87). These findings also indicate that the relationship between purchase situation and relative influence is negatively affected by larger sample sizes (β=−5.20), limited to purchasing respondents only (β=−2.99), and published more recently (β=−5.28). In contrast, the relationship between purchase situation and relative influence is positively affected by higher response rates (β=1.02), representing a larger number of firms (β=2.26) and industries (β=7.72), where data is collected via mail survey (β=1.90). Further, when examining the effects of individual study characteristics we see, for example, that increased study sample size has a consistent negative effect on the relationship between purchase situation and the respective outcome variables. In contrast, increased number of firms included in the study has a consistent positive effect on the relationship between purchase situation and the outcome variables.

In sum, the combined findings of the moderator analysis suggest that the examined study characteristics influence between-study variation between the antecedent and outcome variables. They also suggest that the within-study estimated effect sizes likely are a function of and should be considered in light of the study designs that produce them (Brown et al., 1998).

6. Discussion and directions for future research

One benefit of meta-analysis is its ability to identify and/or clarify unresolved questions in an area of study. To that end, our study contributes in the following ways.

(1) As previously discussed, over the years the buyclass schema has received mixed conceptual and empirical support, resulting in a number of critics. However, contrary to these criticisms, our results provide general support for buyclass as a predictor of buying center size (r'=.47), lateral involvement (r'=.26), vertical involvement (r'=.26), and relative influence (r'=.29)—all significant at $p\leq.05$. At the same time, however, we also find support for the proposition that buyclass alone may be insufficient for understanding important distinctions in buying center structure and involvement. Instead, our findings point to purchase importance, and to a lesser extent purchase uncertainty and complexity, as equally compelling explanatory variables.
(2) The results of our meta-analysis refute allegations that prior studies examining the effects of purchase situation on buying center involvement are inconclusive. Rather, overall our results indicate that findings surrounding the relationships between the nature of the purchase situation and lateral involvement, vertical involvement and relative influence in buying centers are generally robust across studies.
(3) In contrast, our findings support the proposition that, in sum, prior works examining the effect of the purchase situation on buying center formalization and centralization are inconclusive. Within this meta-analysis, these variables generally exhibit the weakest correlations, with all relationships found to be either non- or only marginally significant. These findings, coupled with the concerns expressed by

earlier researchers, point to the need for new, carefully designed research to investigate these important relationships.

A second benefit of meta-analysis is the ability to potentially identify systematic sources of differences across studies within an examined body of knowledge. When such systematic sources are found, future work in the field can be more effectively designed to either eliminate or control for these potential biasing influences. In this regard, our study makes the following contributions.

(1) In sum, our findings indicate the seven examined study design characteristics likely bias effect sizes systematically (regression r-squares ranging from .42 to .87), thus potentially creating confounds with explanatory variables. These findings are reflective of the concerns expressed by several other researchers (cf. Lynch, 1999) over the external validity of study results where background factors are assumed irrelevant.
(2) Of particular interest are the following *select* findings related to the moderator analysis.

    (a) The inconsistency across studies examining the effect of purchase situation on buying center formalization appear to be caused, at least in part, by the number of firms represented in the studies and the method of data collection used. Our findings indicate that in studies examining three or more firms the strength of the relationship between purchase situation and buying center centralization were negatively impacted. One possible explanation is that buying center centralization is dependent on firm characteristics, and thus is difficult to generalize across firms. Similarly, data gathering via mail survey also weakened the relationship between these two constructs. One implication of this finding is the need for the use of multi-methods of data collection in future studies designed to measure the relationship between these two constructs.
    (b) With regard to inconsistent relationships between purchase situation and buying center formalization, the size of the sample and the type of respondent used, along with the number of industries included in the study all seem to significantly influence between study results. More specifically, the relationship between these two constructs is negatively influenced in studies with larger sample sizes and/or limited to purchasing respondents only, and positively influenced in studies where a larger number of industries are included.

We believe that (1) the inconsistencies across findings related to the influence of the purchase situation on buying center centralization and formalization, combined with (2) the robust results of our moderator analysis, provide the most worthwhile directions for future research.

First, our findings indicate that prior attempts to examine the relationship between the nature of the purchase situation and buying center structure are inconclusive. This may be due, in part, to divergent study design characteristics—including some of the characteristics examined here, as well as potential others. One way to test this in future research would be to specify/model expected background variables and then test for the asserted pattern of interactions. Another approach would be for researchers to test the robustness of findings by systematically varying one or more of the domains of their study through replication.

An alternative explanation of our moderator results may lie in the inappropriate nature of prior assumptions. For example, prior studies assume a linear relationship between purchase situation and buying center structure. It is possible, however, that this relationship is nonlinear. Where, for example, levels of formalization and/or centralization increase as levels of purchase importance, complexity, and uncertainty increase to some critical level. Thereafter, levels of formalization and/or centralization begin to decrease as levels of purchase importance, complexity, and uncertainty increase further. Future research might examine these potential relationships.

7. Limitations

Notwithstanding the contributions of the study reported here, meta-analyses share certain limitations that should be considered. First, similar to prior work, our results are constrained by the methods used in the original empirical studies. Also, as is customary in meta-analysis, we assume that the effect sizes accumulated from the original works are from a homogeneous population; and any cause of heterogeneity can be distinguished and segregated. This often is not the case. Therefore, our reported effect sizes are presented as averages and should be viewed accordingly.

Second, the basis of this investigation was limited to published research. Although an extensive effort was made to uncover unpublished manuscripts, no unpublished manuscripts were identified. This potentially could bias the obtained estimates.

Third, while meta-analysis proponents believe it is appropriate to test effects from just a few studies, others remain cautious. Therefore, given the small number of studies related to some of the specific variable pairs, we urge readers to be judicious when drawing conclusions from these particular individual results. In addition, one cell in the moderator analysis had fewer than the minimum desirable number of study effects for bootstrap regression (8 vs. 10). Hence, caution in interpreting the moderator results related to

this cell (purchase situation/buying center centralization) is warranted.

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